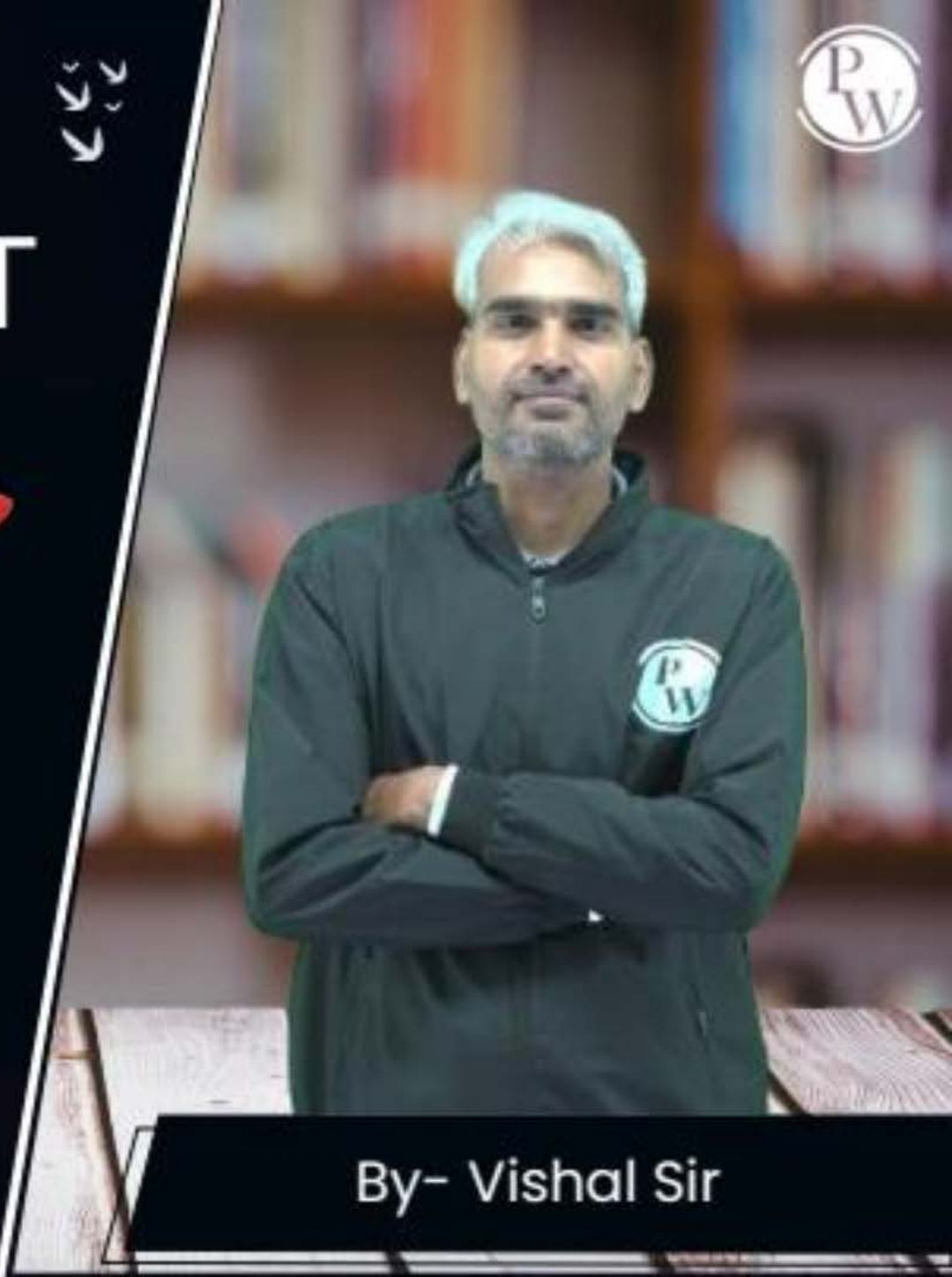
Computer Science & IT



Database Management System

Transaction & concurrency control

Lecture No. 10



By- Vishal Sir

Recap of Previous Lecture



Topic Practice questions on View serializable schedule



Slide

Topics to be Covered



- **Topic** Concurrency control protocols
- **Topic** Simple use of shared and exclusive locks
- **Topic** Two phase lock (2PL)



Topic : Concurrency Control Protocols

- Concurrency Control Protocols are responsible for avoiding the execution of non-serializable schedules.
  - * Concurrency Control protocols will never allow a non-serializable schedule
- Concurrency Control Protocols are suggested to ensure serializability
  - * If schedule is allowed to execute by the Concurrency Control Protocol, then the schedule is a serializable schedule



Topic : Concurrency Control Protocols

- There are two types of concurrency control protocols.

1. Lock based Protocols
   - Two Phase Locking Protocol (2PL)
     - (i) Basic 2PL
     - (ii) Strict 2PL
     - (iii) Conservative 2PL
     - (iv) Rigorous 2PL
2. Time-stamp based Protocols
   - (i) Basic Time stamp ordering Protocol.
   - (ii) Thomas write Time-stamp ordering Protocol



Topic : Lock Based Protocols

- Lock is a variable associated with a data item that describes the status of data item with respect to possible operation that can be applied to it.

There are two types of locks used,

1. Shared LOCK [S]: It is read only lock.

2. Exclusive LOCK [X]: It is read as well as write lock.



Topic : Lock Based Protocols

Lock is a variable associated with a data item that describes a status of data item with respect to possible operation that can be applied to it.

There are two types of locks used,

1. Shared LOCK [S]: It is read only lock.

If transaction wants to perform only "read" op^{n} on a dataitem then transaction will request for a shared lock on that dataitem

T

R(A)

Not allowed because transaction T does not have Permission to Read 'A'

Requesting for a shared lock on dataitem "A"

T

S(A) ← let granted

R(A) ~{Allowed}

W(A) ×{Not allowed}

R(B) ×{Not allowed}

If transaction request for a lock and it is denied, then, transaction will wait for time out "Period and will request again



Topic : Lock Based Protocols

Lock is a variable associated with a data item that describes a status of data item with respect to possible operation that can be applied to it.

There are two types of locks used,

2. Exclusive LOCK [X]: It is read as well as write lock.

Transaction can request for an Exclusive lock on dataitem 'A' only if transaction wants to perform at least one write Operation on dataitem 'A'

Requesting for an Exclusive lock on dataitem A

T

X(A) ← let granted

W(A) ✓ {Allowed}

T

W(A)

denied

S(B) ← let granted

R(A) ✓ {Allowed}

R(B) ✓ {Allowed}

W(B) ✗ Not allowed

R(C)/W(C) ✗ Not allowed



Topic : Lock Compatibility Table

Lock requested by transaction T_j

data item (A)	S	X
S	allowed	Not allowed
X	Not allowed	Not allowed

Lock held by transaction T_i

Note: To ensure serializability a non-serializable schedule must not be allowed to execute

Topic : Simple use of shared and exclusive locks

→ i.e, No restriction { i.e, dataitems can be locked / Unlocked in any order at any time }

Consider the following schedule (S)

T_1	T_2
X(A)	
R(A)	
W(A)	
Unlock(A) → U(A)	
	S(A)
	R(A)
	S(B)
	R(B)
	U(A)
	U(B)
X(B)	
R(B)	
W(B)	
U(B)	

given schedule is a non-serializable schedule.

⇓

But it is allowed to execute using "Simple use" of Shared & Exclusive lock

∴ Simple use of shared (S) and Exclusive (X) lock does not ensure serializability.

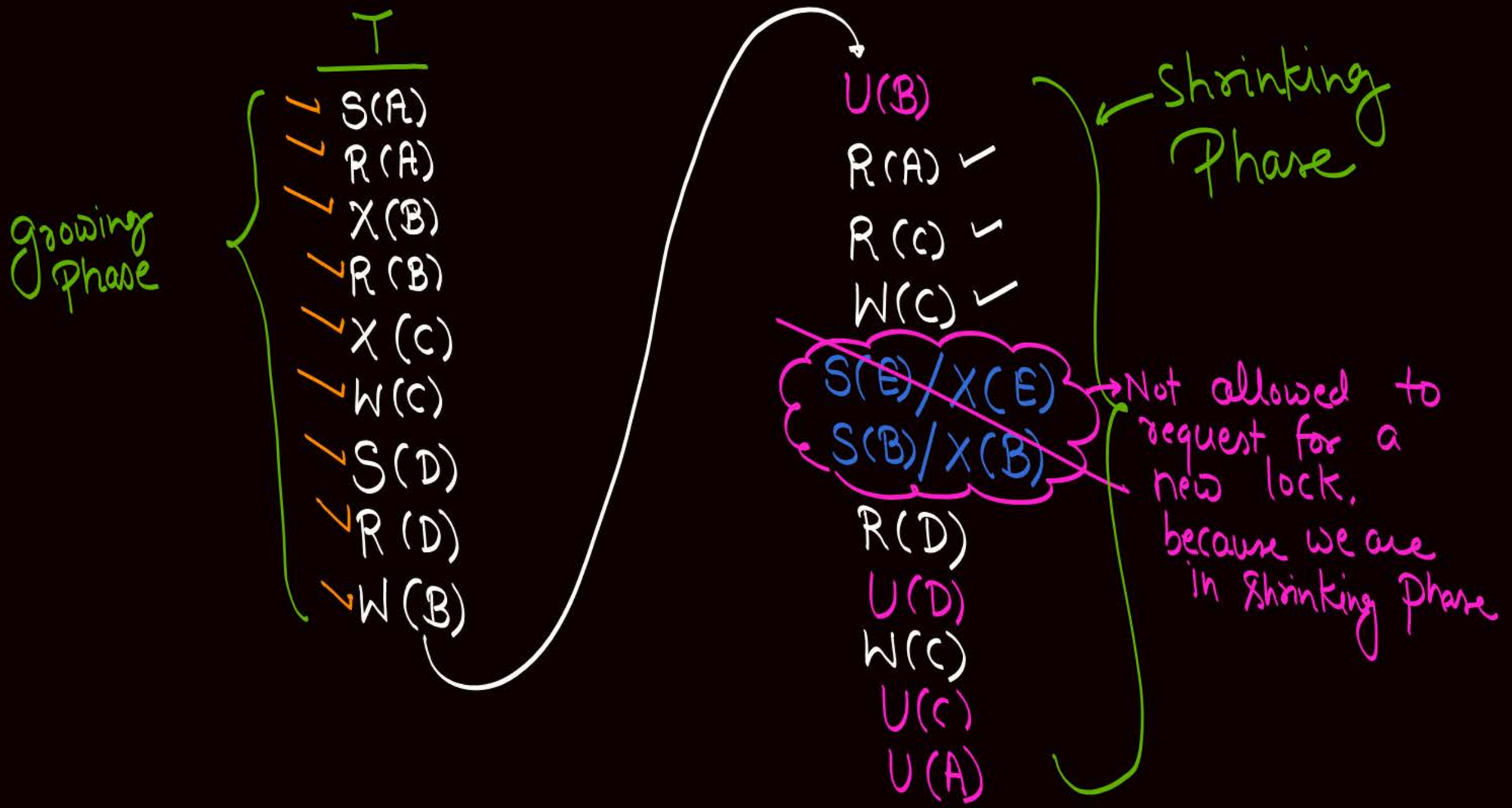


Topic : Basic Two Phase Locking Protocol

There are two phases in Two phase locking protocol

1. Growing phase / Lock acquiring phase
2. Shrinking phase / Lock releasing phase

In Two Phase Locking Protocol, transaction T is allowed to request for a lock only if transaction T has not performed any unlock operation.


T
S(A)
R(A)
X(B)
R(B)
X(C)
W(C)
S(D)
R(D)
W(B)
Growing Phase
U(B)
R(A)
R(C)
W(C)
S(E)/X(E)
S(B)/X(B)
R(D)
U(D)
W(C)
U(C)
U(A)
Shrinking Phase
Not allowed to request for a new lock, because we are in Shrinking Phase

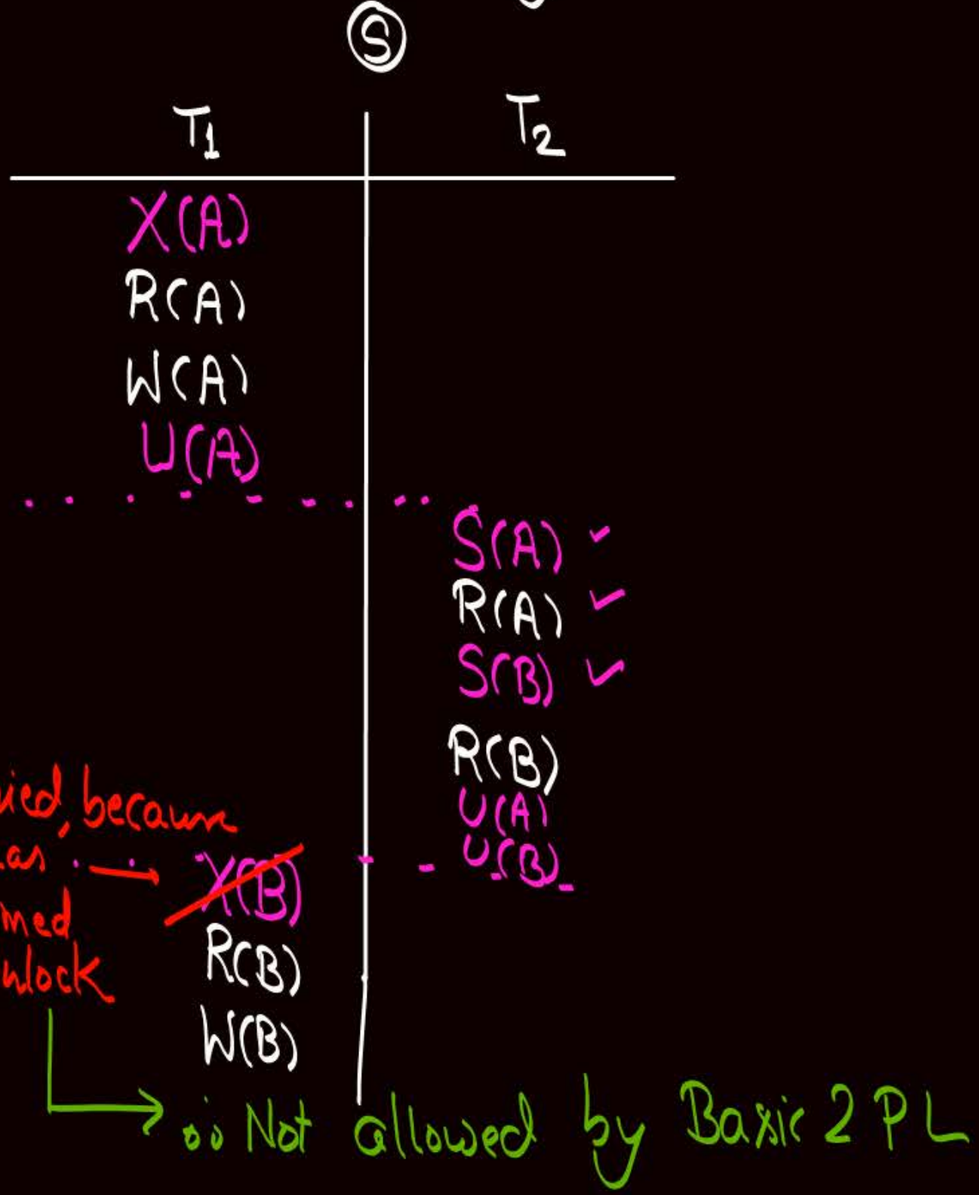
* Check whether the following Schedule is allowed to execute using Basic 2PL or not ?

(S)

T_1	T_2
X(A)	
R(A)	
W(A)	
	S(A) ← it will be denied
	R(A)
	R(B)
R(B)	
W(B)	

↳ ∴ Not allowed by Basic 2PL

* Check whether the following Schedule is allowed to execute using Basic 2PL or not ?



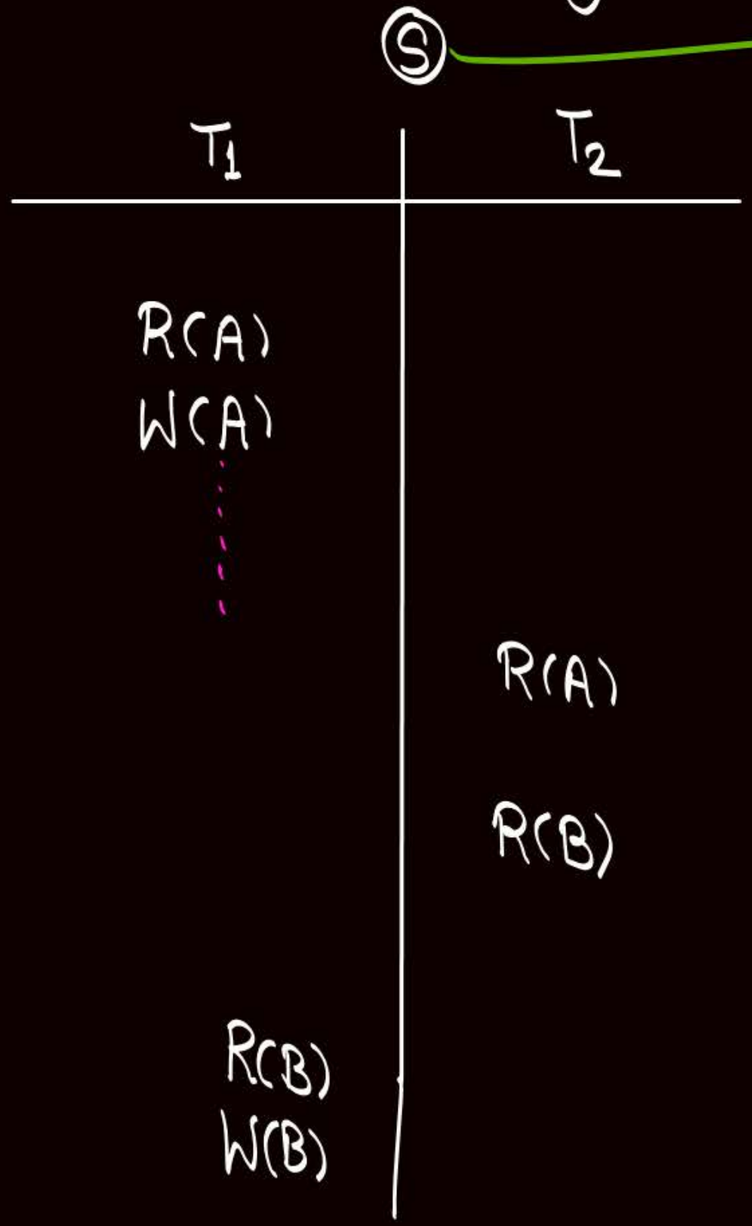
* Check whether the following Schedule is allowed to execute using Basic 2PL or not ?

(S)

T_1	T_2
X(A) ✓	
R(A)	
W(A)	
X(B) ✓	
U(A) ✓	
	S(A) ✓
	R(A)
	~~S(B)~~ ← denied, because 'B' is locked by T_1 in exclusive mode
	R(B)
R(B)	
W(B)	

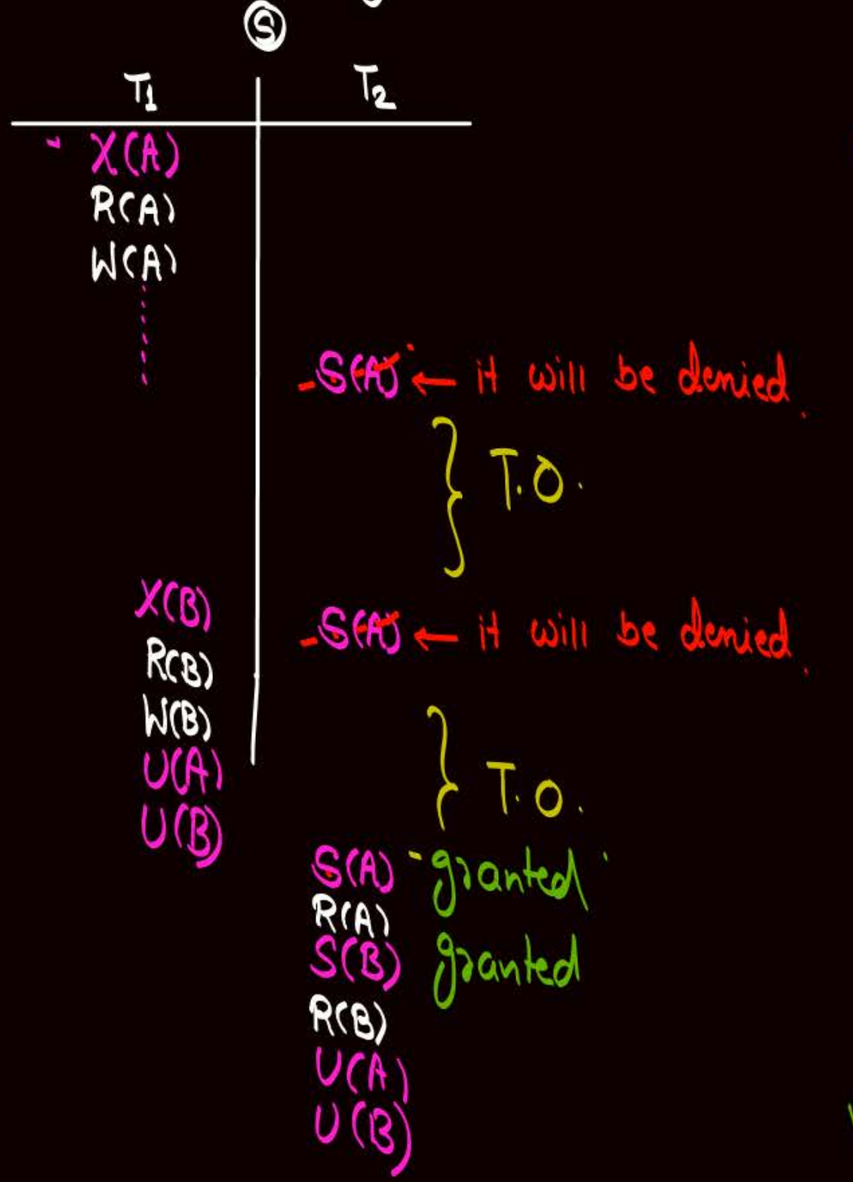
∴ Not allowed by Basic 2PL

* Check whether the following Schedule is allowed to execute using Basic 2PL or not ?



It is a non-serializable schedule & Not allowed by Basic 2PL

* Check whether the following Schedule is allowed to execute using Basic 2PL or not ?



* Using Basic 2PL, this may be the execution order
* But it is not the same Schedule as given Schedule because the order of opn differ

Q:- Check whether the following schedule is allowed using Basic 2PL or not?



T_1	T_2
$X(A)$	
$R_1(A)$	
$W_1(A)$	
$X(B)$	
$U(A)$	
	$S(A)$
	$R_2(A)$
$R_1(B)$	
$W_1(B)$	
$U(B)$	
	$S(B)$
	$R_2(B)$
	$U(A)$
	$U(B)$

Precedence graph



Acyclic
∴ C.S.S.

It is a Conflict serializable schedule and we can observe that the schedule is allowed to execute using Basic 2PL

Q: Check whether the schedule is allowed to execute using Basic 2PL or not?



Precedence graph



acyclic
∴ CSS

T_1	T_2
	X(B)
	R(B)
S(A)	
R(A)	
S(B)	
R(B)	
	W(B)

Until transaction T_2 performs all oph on dataitem 'B' it will not unlock (B)

It is a Conflict Serializable Schedule

But it is not allowed using Basic 2PL

denied → S(B)

∴ Not allowed by "Basic 2PL"

Q:. Check whether the schedule is allowed to execute using Basic 2PL or not ?

Precedence graph ⇒ 

Cyclic
∴ Not a C.S.S.

but $S \equiv T_1 \rightarrow T_2$ ∴ S is Serializable.

Schedule is not a Conflict Serializable Schedule (But Serializable)
↓
and it is not allowed by Basic 2PL

T_1	T_2
X(B)	
R(B)	
	~~X(B)~~ ← denied
	W(B)
W(B)	
	W(B)

∴ Not allowed by Basic 2PL

Note:- ① If schedule is allowed to execute using Basic 2PL then schedule is guaranteed to be a Conflict Serializable schedule (Converse of the statement need not be true)

{ i.e. Basic 2PL ensure Conflict Serializability }

i.e. if schedule is not a C.S.S., then it will never be allowed to execute using Basic 2PL



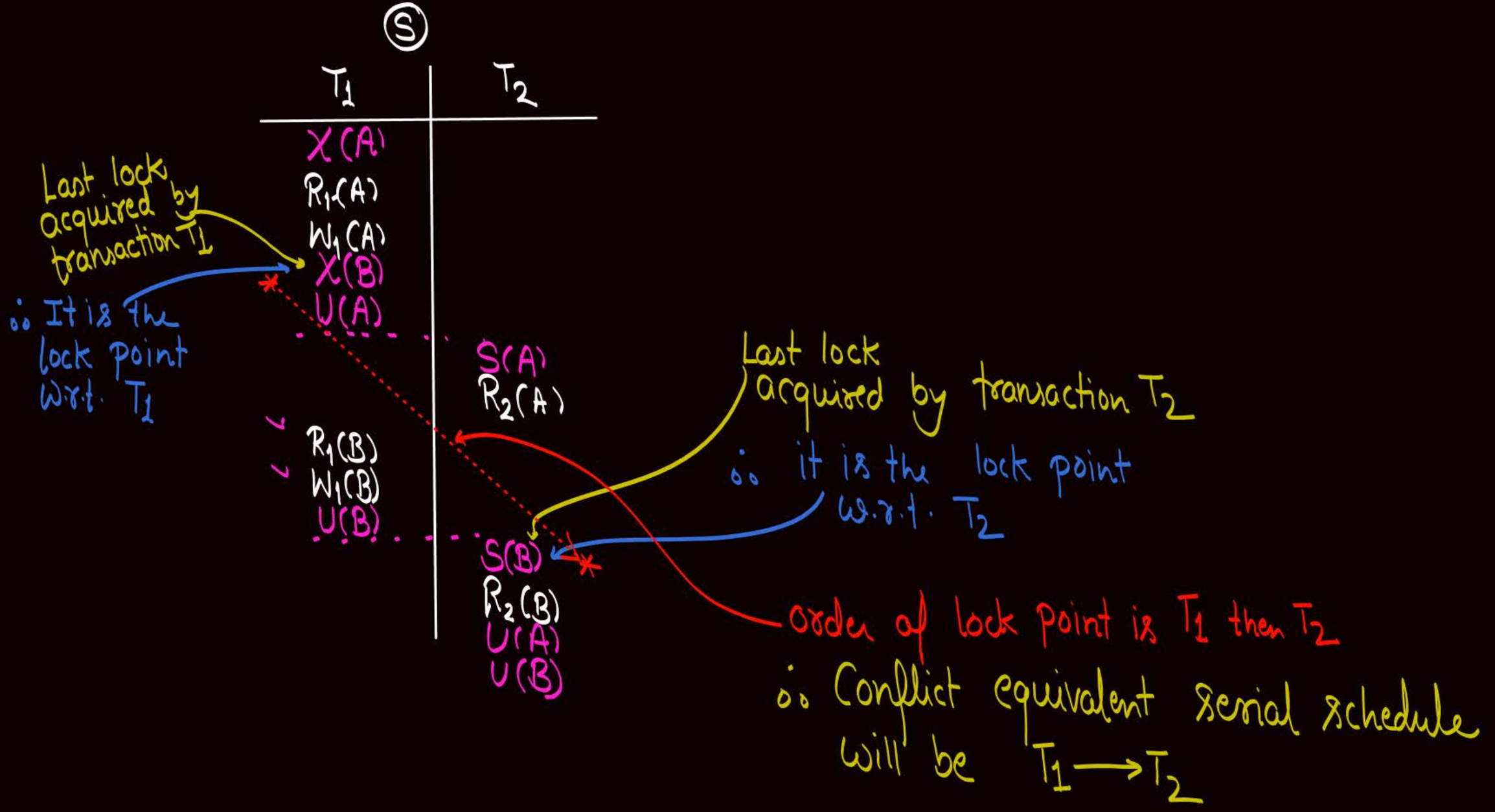
Topic : Basic Two Phase Locking Protocol

1. 2PL ensures serializability, i.e., a non-serializable schedule is never allowed to execute using 2PL.
2. If schedule is allowed to execute using 2PL, then schedule is conflict serializable schedule, but every conflict serializable schedule need not be allowed to execute by 2PL.
3. If schedule is not a conflict serializable schedule, then that schedule is not allowed to execute using 2PL.
4. If schedule is allowed to execute using 2PL, then schedule is conflict serializable schedule and conflict equivalent serial schedule can be given by order of lock points.

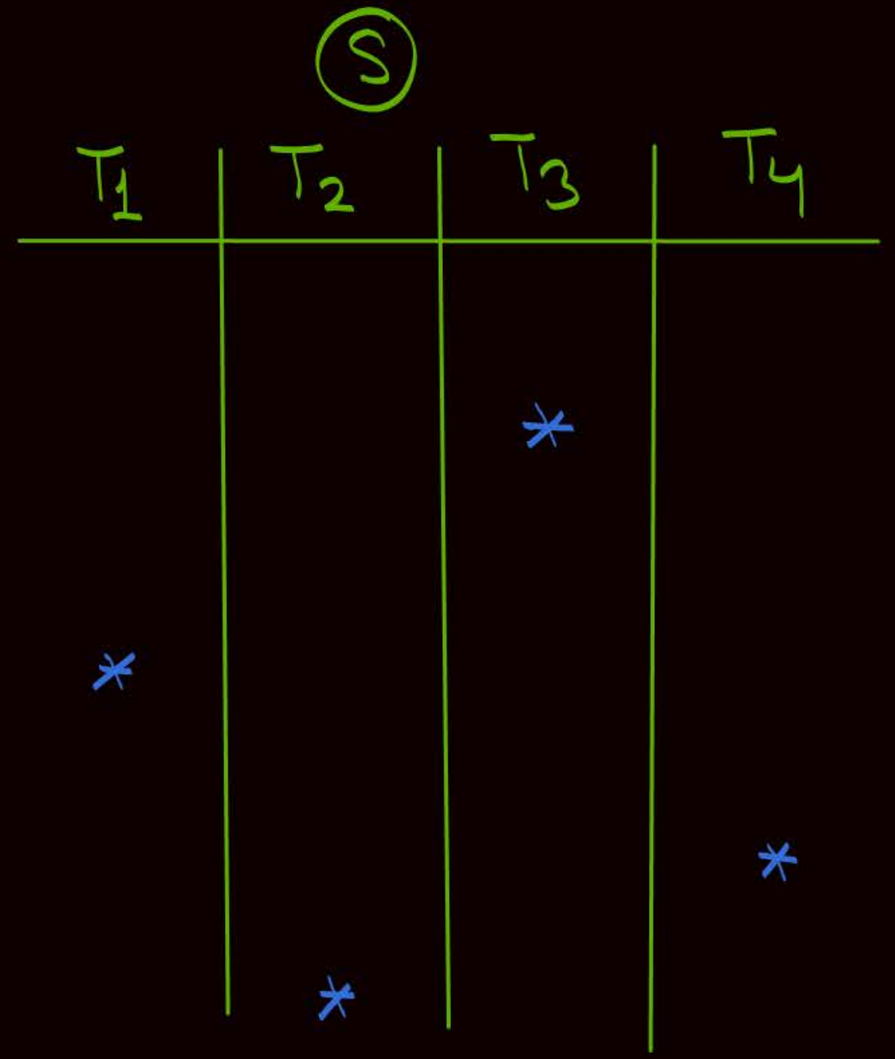


Topic : Lock Point

The Point at which the growing phase ends, i.e., the point at which transaction takes the final lock it needs to carry on its work. is defined as lock point of that transaction


S
T1
T2
X(A)
R1(A)
W1(A)
X(B)
U(A)
R1(B)
W1(B)
U(B)
S(A)
R2(A)
S(B)
R2(B)
U(A)
U(B)
Last lock acquired by transaction T1
∴ It is the lock point w.r.t. T1
Last lock acquired by transaction T2
∴ it is the lock point w.r.t. T2
order of lock point is T1 then T2
∴ Conflict equivalent serial schedule will be T1 → T2

4



Let Schedule S is allowed by basic 2PL, and '*' represent the lock point in the corresponding transaction, then Conflict equivalent Serial Schedule will be

$T_3 \longrightarrow T_1 \longrightarrow T_4 \longrightarrow T_2$



2 mins Summary

- Topic: Concurrency control protocols
- Topic: Simple use of shared and exclusive locks
- Topic: Two phase lock (2PL)

Slide

THANK - YOU